Aaditya Singh Rawat

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EDUCATION

Vellore Institute of Technology Sept 2023 – Present
Bachelor of Technology, Major in Computer Science and Engineering
Cumulative GPA: 8.82

TECHNOLOGIES AND TOOLS

- **Languages:** Java, Python, JavaScript, SQL, HTML, CSS
- **Frameworks & Libraries:** React.js, Node.js, Express.js, Tailwind CSS, Mongoose
- **Databases & Tools:** MongoDB, GitHub, VS Code

PROJECTS UNDERTAKEN

DispatchFlow – Asynchronous Notification Engine
High-Throughput Backend Notification System

- Built a backend service that processes and sends notifications in the background, preventing the main application from freezing or slowing down when external email services respond slowly.
- Implemented a task queue system to manage notification delivery, automatically retrying failed messages and preventing duplicate requests from being processed twice.
- Added input validation checks on incoming requests to catch formatting errors early before they reach the processing stage.
- **Tech Stack:** Node.js, JavaScript, Express, BullMQ, Redis, Docker.

QueryOptima – Database Performance & API Analytics Dashboard
Full-Stack Analytics Dashboard

- Engineered a Node.js and Express backend service designed to process web requests and connect reliably to a MongoDB cloud database.
- Developed Express.js server routes and controllers to process incoming client requests and return formatted JSON data.
- Integrated MongoDB using Mongoose schemas to safely store, update, and retrieve application data.
- Implemented centralized error-handling middleware and used dotenv environment variables to keep database credentials secure.
- **Tech Stack:** Node.js, Express.js, MongoDB, Mongoose, JavaScript.

CERTIFICATIONS

- AWS Technical Essentials, AWS
- Advanced Soft Engg and AI Foundation – MPOnline Limited
- IoT Markets and Security, Coursera (University of Colorado Boulder)

CO-CURRICULARS

- **Lead, Technical Team – Pahadi Club, VIT Bhopal:** Led the technical team across club events, managing infrastructure setup and troubleshooting to ensure smooth execution of cultural programs.
- **National Service Scheme (NSS), VIT Bhopal:** Participated in a plantation drive that planted 10,000+ trees, contributing to a major environmental sustainability initiative.

RESEARCH PRESENTATION

- **Hand Gesture Recognition and Prediction for Accessible Communication Powered by Deep Learning and Computer Vision** – Paper presented at the 17th International Conference on Science & Innovative Engineering (ICSIE 2026), April 2026, Chennai, India, in association with Manipal University College, Malaysia.

CODING PROFILE

- LeetCode